

Strategy Report: Unlocking \$460M in Hidden Revenue

Portfolio Optimization via Hybrid Machine Learning

1. Executive Summary

Traditional credit scoring suffers from "Conservative Bias," rejecting borrowers in sub-prime segments (Grades D/E) who exhibit Prime-level risk profiles. This analysis of **2.26 Million records** identifies a strategy to safely recover this lost yield.

\$460,071,296	\$2.46 Billion	153,307
UNLOCKED ANNUAL REVENUE	RECOVERED LOAN VOLUME	HIDDEN GEMS IDENTIFIED

2. Technical Architecture

To ensure enterprise-grade scalability, the project utilized **Memory Downcasting** and **Histogram-based XGBoost**. The dataset was balanced using **SMOTE** to ensure high sensitivity to default patterns.

Model Performance: 91% Accuracy | 83% Recall (Catching Defaults) | 76% Precision.

3. Strategic Recommendations

- **Transition to 'Behavioral Alpha':** Implement a second-look engine for Grade D/E applicants.

- **Dynamic Pricing:** Offer tiered interest rates based on model-predicted probability of default.
- **NIM Expansion:** Utilize identified high-yield, low-risk segments to expand Net Interest Margins.